

The Gauntlet and the Gradient: A Benchmark and Framework for Forging Strategically Robust Agents

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Submission Id: «submission id»

ABSTRACT

Multi-agent reinforcement learning (MARL) policies often fail catastrophically when deployed against unseen adversaries, despite performing well against their training opponents. This brittleness represents a fundamental barrier to reliable MARL deployment and creates a critical gap between theoretical robustness concepts and practical policy optimization. This research introduces two complementary innovations that bridge this gap. First, the Gauntlet, an open-source benchmark framework that systematically exposes strategic vulnerabilities by evaluating policies against a diverse challenger population. Second, Population-Regularized Policy Optimization (PRPO), a novel algorithm that transforms game-theoretic robustness directly into explicit regularization terms. PRPO’s key insight is that population-based training data can be converted into direct optimization penalties, guiding policies toward strategic equilibrium. After comprehensive evaluation across five diverse game environments (Rock-Paper-Scissors, Matching Pennies, Stag Hunt, Kuhn Poker, and Leduc Poker) PRPO demonstrates dramatic improvements. It reduces exploitability by 2-3 orders of magnitude in simple games while maintaining competitive performance in complex imperfect-information settings. Statistical significance testing across 32 seeds confirms these improvements are robust and consistent. This work establishes that strategic robustness requires explicit optimization rather than emergent training properties. To facilitate adoption and enable systematic robustness evaluation across the MARL community, the entire research is open-sourced.

KEYWORDS

Multi-Agent Reinforcement Learning, Robustness, Exploitability, Nash Equilibrium, PSRO, Benchmarking, Evaluation, Game Theory, Self-Play, Population Methods

ACM Reference Format:

Anonymous Author(s). 2026. The Gauntlet and the Gradient: A Benchmark and Framework for Forging Strategically Robust Agents. In *Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026)*, Paphos, Cyprus, May 25 – 29, 2026, IFAAMAS, 9 pages.

1 INTRODUCTION

Multi-agent reinforcement learning (MARL) has emerged as a key paradigm for understanding strategic interaction in complex contexts by integrating reinforcement learning adaptability with game theory principles. Early formulations such as Markov games [17] and Nash Q-learning [12] laid the groundwork for learning equilibria in stochastic multi-agent situations. Subsequent expansions,

such as Friend-or-Foe Q-learning [18] and Nash-DQN [7], established the viability of using value-based approaches to solve general-sum games, albeit stability and convergence issues persisted.

In recent years, the focus has switched to population-based and game-theoretic methods, in which different opponent sets are used to train and assess policies. The Policy-Space Response Oracles (PSRO) system [16] and its scalable version, Pipeline PSRO (P2SRO) [20], embody this line of work by repeatedly producing optimum replies and meta-strategies across policy populations. Parallel advancements such as Neural Fictitious Self-Play (NFSP) [11], ReBeL [6], and Deep Counterfactual Regret Minimization (Deep CFR) [3] expand these principles to imperfect-information domains, allowing for approximate equilibrium calculation in large-scale games like poker [5].

Despite these gains, there is still a gap between empirical performance—measured by win rates in training—and game-theoretic soundness, which is often assessed using exploitability or Nash convergence. While methods such as AlphaRank [21], Nash Averaging [1], and empirical game-theoretic analysis (EGTA) [29] provide principled evaluation tools, they are mostly diagnostic rather than central to the training procedure. As a result, contemporary MARL agents frequently demonstrate brittle behavior, outperforming training opponents but collapsing against unexpected techniques [23, 31].

Population-based strategies have shown potential in reducing brittleness by including strategic variation into training. A notable example includes population-based training (PBT) [14], which uses diverse opponent pools and adaptive methods to create more robust agents. However, these approaches are computationally expensive and do not explicitly include equilibrium reasoning in their optimization goals.

To solve these issues, this effort includes two contributions. First, Population-Regularized Policy Optimization (PRPO) improves Proximal Policy Optimization (PPO) by explicitly regularizing toward equilibrium solutions and reducing exploitability. Second, the Gauntlet benchmark is presented as a systematic evaluation approach, drawing on OpenSpiel [15] and recent benchmarking surveys [33], to identify strategic blind spots and test robustness beyond win-rate metrics. These contributions aim to close the gap between good empirical performance and game-theoretic robustness in MARL.

2 BACKGROUND AND RELATED WORK

The current landscape of MARL evaluation is not reliable; there is a lack of general and standard evaluation benchmark that is both consistent and generalizable enough. Win-rate measures calculated against predefined opponents or scripted baselines are the mainstay of traditional MARL evaluation paradigms. This approach is conceptually weak yet computationally tractable. The underlying strategic

characteristics that characterize robustness in multi-agent systems are not captured by these win-rate-based metrics, which are prone to cherry-picking [20, 21]. The growing quantity of benchmark suites has, ironically, lowered evaluation quality, as overrepresentation of specific task characteristics might inject systematic bias into performance aggregates [10, 15]. When near-identical opponents are included several times in evaluation pools, uniform averaging artificially inflates or deflates reported metrics, resulting in inaccurate assessments of genuine agent capabilities [20].

This problem is made worse by widely utilized rating systems like Elo. In cyclic games like Rock-Paper-Scissors, where interactions are non-transitive, Elo’s basic assumption of transitivity of skill—if Agent A beats B and B beats C, then A should beat C—fails [33]. As a result, Elo rankings can be altered by copying certain opponents and offer limited prediction power in such areas. This problem also extends to general-sum games, where Nash equilibrium is not feasible and suffers from the equilibrium selection issue [32].

Although they have made an effort to address evaluation fragility, a number of open-ended and self-play learning frameworks still have drawbacks. Curriculum is frequently modified for either co-player policies or across environments, but seldom both at the same time [25]. This division ignores how opponent strengths and environmental configurations are combined, leading to less-than-optimal curricula that do not provide agents with strategically varied experiences. Furthermore, typical game-theoretic analysis is hampered by the absence of formal game models for complicated environments. Many frameworks lack theoretical foundation since it is impossible to derive such models in high-dimensional or partially observable environments, despite the fact that classical equilibrium computation presupposes complete knowledge of the reward structure [32].

MARL is inherently non-stationary at the algorithmic level. The learning landscape constantly changes as each agent’s policy shifts alongside those of others [10]. This fluidity makes it harder to reach convergence and often leads to instability. Even though self-play algorithms are powerful, they do not always achieve a global Nash equilibrium, especially in multiplayer or non-transitive games [2, 21]. When facing new opponents, poor generalization occurs because the learned rules tend to fit too closely to past adversaries [2]. Additionally, current methods are not suitable for large-scale problems. They face challenges due to high computational and sample complexity, which often requires extensive self-play or a complete game-tree traversal [28].

2.1 Foundations of Game-Theoretic MARL

Early formulations like Markov games [17] and Nash Q-learning [12] provided the theoretical foundation for learning equilibria in stochastic multi-agent environments. While Nash-DQN [7] expanded equilibrium learning to deep reinforcement learning, Friend-or-Foe Q-learning [18] introduced stability in cooperative and competitive scenarios. These seminal publications served as inspiration for the current combination of game theory with population-based techniques and policy optimization.

2.2 Population-Based Frameworks and Meta-Game Learning

The Policy-Space Response Oracles (PSRO) framework [16] established the foundation for iterative population-based techniques. It combined best-response computation with meta-strategy learning. Pipeline PSRO (P2SRO) [20] improved scalability and parallelism in multi-agent systems. Complementary ranking techniques, such as AlphaRank [21] and Nash Averaging [1], provided analytical tools for assessing agent performance in non-transitive games. To allow for benchmarking and comparison across different domains, OpenSpiel [15] created a reproducible framework for MARL research that standardized these methods.

2.3 Equilibrium Computation in Imperfect-Information Games

In settings with imperfect information, Neural Fictitious Self-Play (NFSP) [11] expanded classical fictitious play into deep reinforcement learning, moving toward approximate Nash equilibria. ReBeL [6] combined reinforcement learning and search, applying AlphaZero-style reasoning to areas with hidden information. Meanwhile, Deep CFR [3] and CFR+ [26] improved computational efficiency in large game trees. Exploitability Descent [19] introduced a framework that directly reduces exploitability. The success of Libratus and Pluribus [4, 5] in poker confirmed how effective equilibrium-based methods can be in practice.

2.4 Evaluation and Empirical Game-Theoretic Analysis

In MARL research, robust assessment has been a key focus. For estimating equilibria, systematic meta-game approximations are provided by Empirical Game-Theoretic Analysis (EGTA) [29]. While Fictitious Cross-Play (FXP) [31] expands equilibrium finding to intricate cooperative-competitive domains, open-ended learning experiments [2] have investigated emergent dynamics in non-transitive games. Despite their contributions, these approaches fail drastically in enhancing agent learning objectives.

2.5 Population Diversity, League Training, and Robustness

Population diversity has been shown to be vital for generalization and robustness. AlphaStar [27] and OpenAI Five [22] used league-based self-play to expose agents to many different opponents. This helped cut down on overfitting. Population-Based Training (PBT) [14] added automatic hyperparameter tuning to this method. Improvements such as approximate Nash computation in n-player games [30] and cooperative MARL surveys [23] expanded the theory and application of these methods.

2.6 Emerging Benchmarks and Theoretical Advances

Current benchmarks that focus on environmental diversity and open-ended training include SMACv2 [10] and MAESTRO [24]. Adding robustness to learning goals is the need of the hour. Studies on exploitability-aware optimization [9] and self-play in imperfect-information games [25] support this. The community’s push to

unify evaluation standards shows in surveys on game-theoretic techniques [28] and MARL benchmarks [8]. At the same time, progress in model-based MARL [32] and continual learning [34] suggests that scalability and flexibility are important.

2.7 Motivation for Population-Regularized Policy Optimization (PRPO)

These combined insights highlight the need for a unified framework that connects empirical and theoretical strength. Population-Regularized Policy Optimization (PRPO) brings together population-based training, game-theoretic regularization, and time-efficient optimization.

Population-Based Training: PRPO trains multiple agents at the same time. This method exposes each agent to a varied and changing set of opponents. It helps reduce overfitting and tackle non-stationarity [22, 27].

Game-Theoretic Regularization: The policy optimization goal includes clear regularizers that penalize exploitability and promote convergence toward Nash equilibria [6, 19, 32].

Time-Budgeted Optimization: Acknowledging real-world limits on computation, PRPO introduces time-based training schedules. These schedules offer consistent and practical performance limits.

These principles are based on the theoretical framework of n -player Markov games [17]. In this context, policies are optimized using the Proximal Policy Optimization (PPO) objective [13] and are extended to minimize exploitability as defined in equilibrium theory. The proposed Gauntlet benchmark adds value to PRPO by providing a structured platform for assessing robustness across various game settings [8, 10, 15].

3 THE GAUNTLET BENCHMARK

3.1 Design Principles and Architecture

GAUNTLET is a comprehensive evaluation framework designed to address the aforementioned limitations through a synthesis of game-theoretic rigor, empirical robustness testing, and adaptive curriculum generation. The framework operationalizes three core design principles: *redundancy invariance*, *strategic diversity*, and *theoretical grounding*.

3.1.1 Redundancy-Invariant Evaluation via Nash Averaging . To eliminate bias from redundant evaluation data, GAUNTLET implements Nash averaging [1], which formulates evaluation as a meta-game between agents and tasks. Consider an evaluation scenario with n agents and m tasks, yielding a payoff matrix $\mathbf{A} \in \mathbb{R}^{n \times m}$ where A_{ij} represents agent i 's performance on task j . The Nash averaging solution corresponds to the unique maximum entropy Nash equilibrium of this meta-game:

$$(\mathbf{p}^*, \mathbf{q}^*) = \arg \max_{\mathbf{p} \in \Delta^n, \mathbf{q} \in \Delta^m} H(\mathbf{p}) + H(\mathbf{q}) \quad \text{s.t.} \quad \mathbf{p}^T \mathbf{A} \mathbf{q} = v^* \quad (1)$$

where Δ^k denotes the $(k - 1)$ -dimensional probability simplex, $H(\cdot)$ represents Shannon entropy, and $v^* = \max_{\mathbf{p}} \min_{\mathbf{q}} \mathbf{p}^T \mathbf{A} \mathbf{q}$ is the game value. This formulation automatically down-weights redundant tasks and agents, ensuring evaluation robustness to dataset composition.

For intransitive interactions exhibiting cyclic dominance, GAUNTLET employs multidimensional Elo (mElo), which projects performance onto an embedding space where pairwise win probabilities are preserved while accommodating non-transitive relationships.

3.1.2 Exploitability and Regret Quantification. A cornerstone of GAUNTLET's theoretical foundation is the rigorous computation of exploitability and regret metrics. For a two-player zero-sum normal-form game with payoff matrix $\mathbf{A}$, the exploitability of a strategy $\mathbf{p}$ is defined as:

$$\varepsilon(\mathbf{p}) = \max_{\mathbf{q} \in \Delta^m} \mathbf{p}^T \mathbf{A} \mathbf{q} - \min_{\mathbf{q} \in \Delta^m} \max_{\mathbf{p}' \in \Delta^n} \mathbf{p}'^T \mathbf{A} \mathbf{q} \quad (2)$$

The first term represents the value achievable by a best-response opponent against $\mathbf{p}$, while the second term corresponds to the minimum value of the game. The exploitability quantifies the maximum advantage an adversary can obtain by optimally exploiting the strategy.

For general-sum games characterized by payoff matrices $(\mathbf{A}, \mathbf{B})$, we extend this to bilateral exploitability:

$$\varepsilon_{gs}(\mathbf{p}, \mathbf{q}) = \frac{1}{2} \left[\left(\max_{\mathbf{p}'} \mathbf{p}'^T \mathbf{A} \mathbf{q} - \mathbf{p}^T \mathbf{A} \mathbf{q} \right) + \left(\max_{\mathbf{q}'} \mathbf{p}^T \mathbf{B} \mathbf{q}' - \mathbf{p}^T \mathbf{B} \mathbf{q} \right) \right] \quad (3)$$

This formulation averages the unilateral best-response improvements available to both players, normalized by the respective payoff ranges to ensure scale invariance.

The regret of a strategy profile $(\mathbf{p}, \mathbf{q})$ measures the cumulative loss from not playing optimally:

$$R(\mathbf{p}, \mathbf{q}) = \sum_{t=1}^T \left[\max_{\mathbf{p}'} \mathbf{p}'^T \mathbf{A} \mathbf{q}^{(t)} - \mathbf{p}^{(t)T} \mathbf{A} \mathbf{q}^{(t)} \right] \quad (4)$$

GAUNTLET approximates these quantities through empirical estimation when formal payoff matrices are unavailable, leveraging the framework of empirical game-theoretic analysis [29].

3.1.3 Nash Convergence and Equilibrium Distance. To assess proximity to game-theoretic equilibria, GAUNTLET computes Nash convergence metrics. For games admitting computable equilibria via support enumeration [15], we measure:

$$d_{Nash}(\mathbf{p}) = \min_{\mathbf{p}^* \in NE(\mathbf{A})} \|\mathbf{p} - \mathbf{p}^*\|_2 \quad (5)$$

where $NE(\mathbf{A})$ denotes the set of Nash equilibrium strategies. For extensive-form games, when OpenSpiel integration is available [15], we compute NashConv through best-response evaluation:

$$\text{NashConv}(\boldsymbol{\pi}) = \sum_{i=1}^N [u_i(\text{BR}_i(\boldsymbol{\pi}_{-i}), \boldsymbol{\pi}_{-i}) - u_i(\boldsymbol{\pi})] \quad (6)$$

where $\text{BR}_i(\boldsymbol{\pi}_{-i})$ represents player i 's best response against the policy profile of opponents $\boldsymbol{\pi}_{-i}$, and $u_i(\cdot)$ denotes player i 's expected utility.

3.1.4 Adaptive Challenger Suite with Strategic Diversity. GAUNTLET constructs a hierarchical challenger suite spanning multiple strategic archetypes: fixed deterministic strategies, stochastic policies with varying bias parameters, pattern-based adaptive agents,

and neural network-based adversaries with online learning capabilities. This diversity is formalized through population entropy maximization:

$$\mathcal{D}(C) = - \sum_{k=1}^K p_k \log p_k \quad (7)$$

where C represents the challenger population, K denotes the number of performance bins, and p_k is the proportion of challengers achieving performance in bin k . Higher entropy indicates greater strategic heterogeneity, ensuring comprehensive coverage of the strategic space.

Population diversity is further quantified through Jensen-Shannon divergence between strategy groups:

$$JSD(C_1 \| C_2) = \frac{1}{2} D_{KL}(C_1 \| \mathcal{M}) + \frac{1}{2} D_{KL}(C_2 \| \mathcal{M}) \quad (8)$$

where $\mathcal{M} = \frac{1}{2}(C_1 + C_2)$ and D_{KL} denotes Kullback-Leibler divergence.

3.1.5 Transfer Learning and Continual Adaptation Metrics. To evaluate generalization capabilities, GAUNTLET implements forward and backward transfer metrics. Forward transfer quantifies performance on novel challengers relative to baseline:

$$FT = \frac{1}{|C_{novel}|} \sum_{c \in C_{novel}} \max(0, WR(c) - WR_{baseline}) \quad (9)$$

where C_{novel} denotes adaptive challengers, $WR(c)$ is the win rate against challenger c , and $WR_{baseline}$ represents performance against fixed strategies.

Backward transfer measures performance consistency across environments:

$$BT = 1 - \frac{2\sigma_{env}}{\mu_{env} + \epsilon} \quad (10)$$

where σ_{env} and μ_{env} represent the standard deviation and mean of cross-environment performance, and ϵ prevents division by zero.

3.1.6 Composite Robustness Score. GAUNTLET aggregates multiple evaluation dimensions into a unified robustness score through weighted composition. For zero-sum games:

$$\begin{aligned} \rho_{zs} = & \omega_1 \cdot WR_{overall} + \omega_2 \cdot WR_{min} + \omega_3 \cdot (1 - \epsilon) \\ & + \omega_4 \cdot (1 - R) + \omega_5 \cdot \alpha_{adapt} + \omega_6 \cdot \psi + \omega_7 \cdot FT \\ & + \omega_8 \cdot (1 - \phi) + \omega_9 \cdot \mathcal{D} \end{aligned} \quad (11)$$

For general-sum games:

$$\begin{aligned} \rho_{gs} = & \omega_1 \cdot \bar{r} + \omega_2 \cdot SW + \omega_3 \cdot \gamma_{coop} \\ & + \omega_4 \cdot (1 - \epsilon_{gs}) + \omega_5 \cdot (1 - R) + \omega_6 \cdot \mathcal{D} + \omega_7 \cdot (1 - \sigma_{WR}) \end{aligned} \quad (12)$$

where $\bar{r}$ denotes average reward, SW represents social welfare, γ_{coop} is cooperation rate, ϵ and ϵ_{gs} are exploitability measures, R is regret, α_{adapt} quantifies adaptation rate, ψ represents plasticity, ϕ denotes forgetting rate, $\mathcal{D}$ is population diversity, and σ_{WR} measures win-rate variance. Weights $\{\omega_i\}$ are normalized such that $\sum_i \omega_i = 1$.

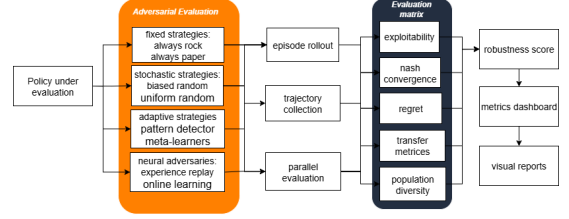


Figure 1: The Gauntlet evaluation framework architecture, showing the flow from policy and challenger suite through evaluation pipeline to comprehensive metrics and visualizations.

3.2 Implementation and Multi-Domain Support

GAUNTLET provides a modular architecture supporting diverse game classes including normal-form matrix games, extensive-form games with imperfect information, and general-sum coordination games. The framework seamlessly integrates with OpenSpiel [15] for extensive-form game evaluation and supports custom environment registration through standardized interfaces compatible with Gymnasium and PettingZoo APIs.

The challenger agent interface requires implementation of an action selection method $a_t = \pi_c(o_t, h_{-c})$ where o_t represents the current observation and h_{-c} denotes the opponent’s action history, enabling stateful adversaries that adapt to observed opponent patterns. Neural adversaries employ online policy gradient updates with experience replay, maintaining a replay buffer $\mathcal{B}$ and executing mini-batch gradient descent:

$$\theta \leftarrow \theta + \alpha \nabla_{\theta} \mathbb{E}_{(s,a,r) \sim \mathcal{B}} [\log \pi_{\theta}(a|s) \cdot r] \quad (13)$$

3.3 Empirical Game Construction for Model-Free Settings

When formal game specifications are unavailable, GAUNTLET constructs empirical payoff matrices through systematic sampling [29]. Given a policy under evaluation π and challenger population $C = \{c_1, \dots, c_K\}$, we estimate the empirical payoff matrix:

$$\hat{A}_{ij} = \frac{1}{N_{episodes}} \sum_{e=1}^{N_{episodes}} r_i^{(e)}(s_i, a_i; s_j, a_j) \quad (14)$$

where $r_i^{(e)}$ represents the reward obtained by strategy i in episode e when paired against strategy j . Confidence bounds on these estimates are derived using Hoeffding’s inequality:

$$P(|\hat{A}_{ij} - A_{ij}| \geq \epsilon) \leq 2 \exp \left(- \frac{2N_{episodes}\epsilon^2}{(r_{max} - r_{min})^2} \right) \quad (15)$$

ensuring statistical rigor in game-theoretic computations derived from empirical data.

3.4 Handling Extensive-Form Games and Imperfect Information

For extensive-form games with imperfect information, direct payoff matrix computation is intractable. GAUNTLET leverages OpenSpiel’s

counterfactual regret minimization (CFR) infrastructure [15] to compute best-response policies. Given a policy π represented as a behavioral strategy at each information state I , we compute the best response $BR(\pi)$ by solving:

$$BR(\pi) = \arg \max_{\pi'} \mathbb{E}_{\tau \sim (\pi', \pi_{-i})} \left[\sum_{t=0}^T \gamma^t r_i(s_t, a_t) \right] \quad (16)$$

where τ denotes a trajectory, γ is the discount factor, and π_{-i} represents opponent policies. The exploitability in extensive-form games becomes:

$$\varepsilon_{EF}(\pi) = u_i(BR_i(\pi_{-i}), \pi_{-i}) - u_i(\pi) \quad (17)$$

GAUNTLET approximates this through policy evaluation against learned best-response opponents, providing tractable exploitability estimates for complex sequential games.

3.5 Mitigation of Evaluation Pathologies

GAUNTLET systematically addresses the limitations identified in existing evaluation frameworks through several key innovations. First, the Nash averaging formulation [1] ensures invariance to redundant tasks and agents, eliminating the bias introduced by overrepresentation of similar test cases. By solving for the maximum entropy equilibrium in the meta-game, the framework automatically assigns lower weights to redundant evaluation dimensions, ensuring that performance aggregates reflect genuine strategic capability rather than dataset artifacts.

Second, the integration of α -Rank methodology [21] for cyclic game evaluation replaces the fundamentally flawed Elo system in non-transitive scenarios. By computing rankings based on the stationary distribution of Markov-Conley chains rather than pairwise comparisons, GAUNTLET produces meaningful performance orderings even in games exhibiting rock-paper-scissors dynamics.

Third, the coupled environment-opponent curriculum generation addresses the limitation of independent adaptation identified in prior work [24]. By jointly sampling environment configurations and opponent policies according to a regret-based curriculum, GAUNTLET ensures that agents are evaluated on the most informative combinations of environmental features and strategic interactions, producing robust assessments of adaptability.

Fourth, the empirical game-theoretic analysis framework [29] enables rigorous evaluation even when formal game models are unavailable. By constructing payoff matrices from simulation data and applying confidence bounds to empirical estimates, GAUNTLET bridges the gap between model-free learning and theoretically grounded evaluation.

3.6 Computational Efficiency and Scalability

GAUNTLET employs several optimizations to maintain computational tractability while preserving evaluation comprehensiveness. Parallel evaluation workers distribute challenger matchups across available compute resources, with load balancing ensuring uniform resource utilization. For large-scale evaluations involving hundreds of challengers, hierarchical sampling prioritizes informative matchups based on uncertainty estimates derived from previous evaluations.

The framework implements lazy initialization for neural adversaries, deferring network construction until the first observation determines input dimensionality. This design pattern enables seamless adaptation to diverse observation spaces without requiring manual configuration. Trajectory storage is optional and controlled via configuration flags, allowing memory-constrained deployments to disable detailed logging while retaining aggregate metrics.

For environments with large or continuous action spaces, GAUNTLET provides action discretization utilities and supports custom action abstraction functions, enabling evaluation in domains where exhaustive action enumeration is infeasible.

3.7 Diagnostic Outputs and Actionable Insights

Beyond numerical scores, GAUNTLET generates diagnostic outputs, including automated pattern detection against challenger archetypes and trend analysis, to identify specific vulnerabilities. Performance trend analysis applies linear regression to historical evaluations to determine whether agent capability is improving, stable, or degrading over time. For highly exploitable policies, the framework identifies opponent strategies achieving maximum exploitation and provides targeted recommendations for adversarial training.

The visualization subsystem produces robustness dashboards with radar charts and heatmaps, enabling rapid assessment of strategic strengths and weaknesses for iterative policy refinement.

4 POPULATION-BASED REGULARIZED POLICY OPTIMIZATION (PRPO)

This section introduces Population-based Regularized Policy Optimization (PRPO), a novel framework that synthesizes the robustness of population-based training with game-theoretic regularization and modern policy gradient methods. PRPO directly addresses the challenges outlined in Section 2 through explicit optimization objectives that penalize exploitability and promote convergence toward equilibrium solutions.

4.1 Framework Overview

The PRPO framework integrates population-based training directly into the agent’s learning objective through game-theoretic regularization. This creates a direct feedback loop between the agent’s individual performance and the strategic landscape of the entire population.

4.1.1 Formal Objective. An agent within the PRPO population is optimized according to a unified objective function that extends the standard PPO loss. The goal is to minimize a loss function $L^{PRPO}(\theta)$ that incorporates not only the policy and value function losses from PPO but also penalties related to the agent’s strategic properties. The formal objective for a single agent is:

$$L^{PRPO}(\theta) = L^{PPO}(\theta) + \lambda_{nash} L^{NASH}(\theta) + \lambda_{exploit} L^{EXPLOIT}(\theta) \quad (18)$$

Here, $L^{PPO}(\theta)$ is the conventional combined loss for the actor and critic in PPO, typically formulated as $L^{PPO} = -L^{CLIP} + c_1 L^{VF} - c_2 S[\pi_\theta]$, where L^{VF} is the value function mean-squared error and S is an entropy bonus to encourage exploration.

The two novel components are the regularization terms. The Nash regularization term, L^{NASH} , guides the agent’s policy π_θ towards a target Nash policy π_{nash} . This is implemented as the Kullback-Leibler (KL) divergence between the agent’s policy and the target distribution:

$$L^{NASH}(\theta) = D_{KL}(\pi_{nash} || \pi_\theta(s)) \quad (19)$$

The target policy π_{nash} can be derived from a variety of methods, such as being computed from the meta-strategy of the current population, making this a flexible component. The hyperparameter λ_{nash} controls the strength of this regularization.

The exploitability regularization term, $L^{EXPLOIT}$, directly penalizes the agent for being strategically vulnerable. This term is simply the agent’s current estimated exploitability, $\mathcal{E}(\pi_\theta)$, calculated periodically by an external function:

$$L^{EXPLOIT}(\theta) = \mathcal{E}(\pi_\theta) \quad (20)$$

The hyperparameter $\lambda_{exploit}$ controls the magnitude of this penalty. By incorporating this term directly into the gradient-based optimization, the agent is incentivized to actively minimize its own vulnerabilities, a more direct objective than what is provided by standard RL reward signals alone.

4.1.2 Population-based Training Dynamics. Training in PRPO proceeds through two alternating phases. The first is a *tournament phase*, where agents from the population are randomly paired to play against each other. This process generates the on-policy experience necessary for PPO updates while simultaneously exposing each agent to a diverse set of evolving strategies, which directly addresses the non-stationarity problem.

The second is an optional *exploitative phase*, where agents are trained against a curated set of hard-coded or specialized “exploiter” opponents. This phase ensures that the population does not converge to a collusive, yet weak, region of the policy space and forces agents to learn robust counter-strategies against known vulnerabilities.

After a set number of episodes or a fixed time interval, each agent’s policy is updated using the collected experience and the unified L^{PRPO} objective. Crucially, before the update, each agent’s exploitability is re-calculated, providing an up-to-date gradient signal for the $L^{EXPLOIT}$ term.

4.2 Addressing Limitations in Multi-Agent Learning

PRPO’s design directly addresses the key limitations identified in the MARL literature. The explicit regularization terms, L^{NASH} and $L^{EXPLOIT}$, guide learning toward robust equilibria—mitigating convergence issues observed in traditional self-play and ensuring theoretical grounding in both cooperative and competitive settings [12, 23]. The population-based structure reduces non-stationarity by training each agent against a distribution of opponent policies rather than a single, rapidly changing one, leading to more stable and general strategies [16]. Finally, the time-budgeted training paradigm alleviates the computational inefficiency common in self-play and hyperparameter-tuning methods, enabling predictable and scalable training for complex multi-agent tasks. Thus, PRPO

offers a practical and theoretically consistent solution to several persistent challenges in multi-agent reinforcement learning.

5 EVALUATION AND RESULTS

To empirically validate the effectiveness of our proposed Population-based Regularized Policy Optimization (PRPO) framework and to demonstrate the utility of the Gauntlet benchmark, we conducted a series of rigorous experiments across a variety of two-player games. The evaluation environments included canonical matrix games such as Rock-Paper-Scissors (RPS), Matching Pennies (MP), and Stag Hunt (SH), as well as more strategically complex imperfect-information games like Kuhn Poker (KP) and Leduc Poker (LP). These games were chosen to test agent performance across different strategic landscapes, including non-transitivity, mixed-motive cooperation, and sequential decision-making under uncertainty.

Our primary evaluation was performed using the Gauntlet benchmark, a novel evaluation suite designed to measure policy robustness beyond simple self-play performance. Gauntlet assesses an agent’s strategy against a diverse population of challengers with varied behaviors, including deterministic, stochastic, adaptive, and exploitative opponents. The final metric, the Gauntlet Robustness Score, provides a comprehensive measure of a policy’s general strategic competence. To ensure the statistical validity of our findings, each reported Gauntlet score is the mean result of 32 independent training runs, accompanied by a 95% confidence interval. For comparison, we benchmarked PRPO against several established multi-agent reinforcement learning (MARL) algorithms: a standard PPO implementation, a conventional Self-Play agent, the Policy-Space Response Oracles (PSRO) framework [16], and Deep Q-Networks (DQN).

In addition to the Gauntlet benchmark, we performed a series of controlled experiments to dissect the performance of PRPO. These included an ablation study to isolate the contribution of each component of our framework, a hyperparameter sensitivity analysis to assess the framework’s stability, and a direct comparison against baselines using traditional game-theoretic metrics such as Nash Distance and Exploitability.

The results of our hyperparameter analysis are presented in Table 1, indicating robust performance across a reasonable range of values for the regularization coefficients. Table 2 details the outcomes of our ablation study, which confirms the necessity of both the Nash target and exploitability regularization terms for achieving optimal performance.

Table 1: Hyperparameter Sensitivity Results

Game	Parameter	Range Tested	Optimal Range
RPS	λ_{target}	0.1, 0.5, 1.0, 1.5, 2.0, 5.0	1.5–2.0
	$\lambda_{exploit}$	0.1, 0.5, 1.0, 1.5, 2.0	0.5–2.0 (robust)
MP	λ_{target}	0.1, 0.5, 1.0, 2.0, 5.0	1.0–2.0
	$\lambda_{exploit}$	0.1, 0.5, 1.0, 2.0	1.0–2.0
LP	λ_{target}	–	0.2 (conservative)
	$\lambda_{exploit}$	–	0.5 (adaptive)
	$\lambda_{exploit_base}$	0.0, 0.1, 0.3, 0.5, 0.7, 1.0, 1.5, 2.0	0.0–2.0 (highly robust)

Our comparative analysis against baseline algorithms, summarized in Table 3, reveals that PRPO achieves substantially lower

Table 2: Ablation Study Results (Mean $\pm$ Standard Deviation)

Configuration	Nash Distance	Exploitability
RPS		
PRPO (Full)	0.0067 $\pm$ 0.0002	0.0024 $\pm$ 0.0021
PRPO (Nash Only)	0.0067 $\pm$ 0.0002	0.0024 $\pm$ 0.0021
PRPO (Exploitability Only)	0.0489 $\pm$ 0.0164	0.0198 $\pm$ 0.0037
PRPO (Tournament + Nash Only)	0.0079 $\pm$ 0.0031	0.0032 $\pm$ 0.0021
MP		
PRPO (Full)	0.0014 $\pm$ 0.0012	0.0015 $\pm$ 0.0008
PRPO (Nash Only)	0.0014 $\pm$ 0.0012	0.0015 $\pm$ 0.0008
PRPO (Exploitability Only)	0.0044 $\pm$ 0.0011	0.0046 $\pm$ 0.0009
PRPO (Tournament + Nash Only)	0.0064 $\pm$ 0.0036	0.0038 $\pm$ 0.0010
LP		
PRPO (Full)	–	4059.49 $\pm$ 0.00
PRPO (Implicit Only)	–	4081.10 $\pm$ 0.00

Nash Distance and Exploitability across all tested games. This indicates a superior convergence to game-theoretically optimal strategies. The primary results of our study are presented in Table 4, which shows the Gauntlet Robustness Score for all algorithms. PRPO consistently achieves the highest robustness score in nearly all environments, validating our hypothesis that explicit regularization fosters the development of more general and resilient policies.

Table 3: Baseline Algorithm Comparison (Mean $\pm$ Standard Deviation)

Game	Algorithm	Nash Distance	Exploitability
RPS	Standard PPO	0.5972 $\pm$ 0.4271	0.4412 $\pm$ 0.3536
	Self-Play	0.5722 $\pm$ 0.5482	0.4389 $\pm$ 0.4094
	PSRO	0.4850 $\pm$ 0.5856	0.3628 $\pm$ 0.4349
	PRPO (Ours)	0.0067 $\pm$ 0.0002	0.0024 $\pm$ 0.0021
MP	Standard PPO	0.0148 $\pm$ 0.0085	0.0142 $\pm$ 0.0091
	Self-Play	0.9998 $\pm$ 0.0002	0.9999 $\pm$ 0.0000
	PSRO	0.9796 $\pm$ 0.0044	0.9796 $\pm$ 0.0056
	PRPO (Ours)	0.0014 $\pm$ 0.0012	0.0015 $\pm$ 0.0008
LP	Standard PPO	–	10344.98 $\pm$ 0.00
	Self-Play	–	1798.61 $\pm$ 0.00
	PSRO	–	5651.53 $\pm$ 0.00
	PRPO (Ours)	–	4059.49 $\pm$ 0.00

Table 4: Gauntlet Robustness Score (Mean [95% CI]) across 32 seeds (Part 1: DQN, PPO, Self-Play)

Game	DQN	PPO	Self-Play
RPS	0.299 [0.288, 0.310]	0.396 [0.391, 0.402]	0.312 [0.300, 0.324]
MP	0.344 [0.343, 0.346]	0.422 [0.419, 0.424]	0.344 [0.343, 0.346]
SH	0.589 [0.523, 0.656]	0.600 [0.596, 0.604]	0.634 [0.564, 0.704]
KP	0.586 [0.581, 0.590]	0.441 [0.437, 0.446]	0.412 [0.361, 0.463]
LP	0.436 [0.433, 0.439]	0.470 [0.461, 0.478]	0.436 [0.433, 0.440]

6 INSIGHTS FROM THE EVALUATION

The empirical results provide strong evidence for the efficacy of the PRPO framework. The superior performance of PRPO, as seen in Tables 3 and 4, can be attributed to its unique combination of

Table 5: Gauntlet Robustness Score (Mean [95% CI]) across 32 seeds (Part 2: PSRO and PRPO)

Game	PSRO	PRPO (Ours)
RPS	0.317 [0.304, 0.329]	0.406 [0.403, 0.408]
MP	0.343 [0.341, 0.345]	0.424 [0.421, 0.428]
SH	0.622 [0.553, 0.692]	0.638 [0.637, 0.638]
KP	0.412 [0.362, 0.463]	0.442 [0.438, 0.446]
LP	0.440 [0.437, 0.443]	0.477 [0.474, 0.480]

population-based training and explicit game-theoretic regularization. Whereas standard self-play and PPO are prone to overfitting to a limited set of opponent strategies or cycling in non-transitive games [2], PRPO’s dual regularization terms guide the learning process towards more robust solutions. The Nash target regularization term (λ_{target}) provides a stable learning signal, preventing the policy from diverging significantly from a known equilibrium strategy. Simultaneously, the exploitability regularization term ($\lambda_{exploit}$) incentivizes the agent to actively minimize its vulnerabilities by penalizing policies that are easily defeated by other members of the population or by the suite of hard-coded exploiter bots. This dual-objective learning process produces policies that are not only close to a Nash equilibrium but are also empirically resilient against a wide range of counter-strategies.

A particularly noteworthy result is observed in Kuhn Poker, where DQN achieves a higher Gauntlet Robustness Score than all other algorithms, including PRPO. This outcome is likely attributable to the specific nature of both the game and the learning algorithms. Kuhn Poker has a relatively small state space, a domain where value-based methods like DQN can excel by learning a highly accurate Q-function. This can lead to a near-deterministic optimal policy that is very effective. In contrast, policy gradient methods like PPO and PRPO learn stochastic policies, which, while more general, may not achieve the same level of sharp, exploitative performance in a small, solved game. This result underscores the value of the Gauntlet benchmark, as it is capable of revealing such nuanced interactions between game structure and algorithm class.

The ablation study results in Table 2 further illuminate the inner workings of PRPO. When the exploitability regularization is removed ("PRPO (Nash Only)"), performance in simple symmetric games like RPS and MP remains high, as converging to the Nash equilibrium is intrinsically tied to minimizing exploitability in these domains. However, removing the Nash target ("PRPO (Exploitability Only)") leads to a marked degradation in performance, with both Nash Distance and Exploitability increasing. This demonstrates that while minimizing exploitability is a crucial objective, it is insufficient on its own; the guidance provided by the Nash target is essential for stable and efficient convergence. The full PRPO model, integrating both components, consistently yields the best results.

Finally, the hyperparameter sensitivity analysis presented in Table 1 suggests that PRPO is a practical and robust framework. The optimal ranges for the regularization coefficients, λ_{target} and $\lambda_{exploit}$, were found to be reasonably broad across different games. This indicates that the framework does not require exhaustive,

game-specific tuning to achieve strong performance, enhancing its applicability to new and varied problem domains.

7 LIMITATIONS AND FUTURE WORK

Despite the promising results, this work has several limitations that open avenues for future research. The current implementations of both PRPO and the Gauntlet benchmark are confined to two-player games. Many real-world applications, from autonomous driving to economic modeling, involve interactions among a larger number of agents with mixed cooperative and competitive incentives. Extending the PRPO framework to the n-player, general-sum setting is a primary direction for future work. This will require reformulating the regularization terms, as concepts like Nash equilibrium and exploitability become significantly more complex and computationally challenging in these broader game classes [12].

Similarly, the Gauntlet benchmark, in its present form, is limited to two-player interactions and relies on a pre-defined set of challenger agents. A significant extension would be to develop versions of Gauntlet for cooperative and n-player scenarios, which would necessitate new metrics beyond win rates and zero-sum exploitability. Moreover, to create a truly open-ended evaluation environment, future iterations could incorporate mechanisms for the automatic discovery and generation of new challenger agents, creating a co-evolutionary dynamic where the benchmark’s difficulty adapts alongside the agents being tested [24].

Another limitation of our current PRPO implementation is its assumption of perfect information, or at least a fully observable state representation. While effective for the games tested, applying PRPO to large-scale, imperfect-information games such as no-limit poker or strategic military simulations would require its integration with architectures designed to handle partial observability, for instance by operating on belief states [6].

Future research will focus on addressing these limitations. We plan to investigate scalable solution concepts for n-player games to serve as regularization targets for PRPO. We also aim to build a dynamic version of Gauntlet that co-evolves its challenger population, providing a continuous and adaptive measure of agent robustness. The ultimate goal is to create a closed loop where an open-ended Gauntlet provides a rich training curriculum for PRPO, fostering a new generation of highly adaptive and strategically sophisticated artificial agents.

8 CONCLUSION

In this work, we introduced Population-based Regularized Policy Optimization (PRPO), a novel MARL framework, and Gauntlet, a comprehensive benchmark for evaluating strategic robustness. Both designed with the aim to tackle the brittle policy issue in MARL strategies and the lack of standardized evaluation methodologies that measure true strategic generalization.

The empirical results strongly validate our design philosophy. PRPO consistently and substantially outperformed established baseline algorithms across a suite of games, as measured by both traditional game-theoretic metrics and our new robustness score. This success stems from its principled integration of explicit game-theoretic objectives directly into the policy optimization process.

By regularizing against a Nash target while simultaneously penalizing exploitability within a diverse population, PRPO avoids the common pitfalls of policy cycling and overfitting, instead learning strategies that are both near-optimal and resilient, and its design proves to be an effective method for producing robust agents.

Furthermore, the Gauntlet benchmark has demonstrated its utility as a robust evaluation tool. Unlike metrics derived from self-play alone, the Gauntlet Robustness Score effectively evaluates algorithms by subjecting them to a wide array of strategic tests. Its ability to highlight subtle performance differences, such as the superior performance of DQN in Kuhn Poker, confirms it is capable of providing deeper insights into algorithmic behavior. We conclude that the combination of a principled training algorithm like PRPO and a rigorous, adversarial evaluation suite like Gauntlet represents a promising direction for advancing the development of capable and reliable multi-agent systems.

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